Ryan Kim

New Jersey | 201-245-8979 | rkydin404@gmail.com | linkedin.com/in/ryandki | rkydin.github.io | U.S. Citizen
Open to relocation and nationwide travel opportunities

Technical Skills

Mechanical Design: SolidWorks; SolidWorks Simulation; Abaqus; Siemens NX; GD&T; Tolerance Stack-Ups; DFM; DFA; FEA; 2D Drawings; BOM Management

Manufacturing: Prototype Assembly; Fixture Design; Root Cause Analysis; ISO 9001

Laboratory: Mechanical Testing; Failure Analysis; Data Analysis; Instron Testing; Optical Microscopy; GLP; Oscilloscopes; DMM

Programming: Python; MATLAB; C++

Experience

Volta Space Technologies — April 2026 – July 2026
Mechanical Design Intern — *Broomfield, CO*

- Designed a CAD enclosure housing a laser assembly and its associated optical fixtures for shipment and testing, supporting delivery to the **European Space Agency (ESA)**.
- Gained hands-on exposure to electromechanical assembly, including **soldering** and **cable harnessing**, through hardware build support.
- Developed **SolidWorks** CAD models and mechanical hardware for internal and electro-optical support applications.
- Designed optical-mount fixtures and supported hardware development for future **thermal vacuum (TVAC)** testing.
- Maintained drawing and BOM revision control through **ECOs** and **redlines**, ensuring documentation accuracy across engineering and technical reporting in Microsoft 365.

Robomekanics — Nov 2025 – Feb 2026
Mechanical Engineer (Contract, 3 months) — *Teterboro, NJ*

- Reduced part count by **15%** across pallet-handling robotics assemblies by redesigning and merging components during the metric-to-inch CAD conversion, cutting fabrication and assembly labor.
- Converted legacy metric CAD assemblies into inch-based production models and **ASME Y14.5**-compliant drawings within an **ISO 9001** quality program.
- Resolved fit-up, interference, and failure issues on sheet-metal and welded assemblies by performing **tolerance stack-ups** and revising geometry with machinists and fabricators.
- Designed welding/assembly fixtures and updated build procedures with **redlined drawings**, improving fabrication repeatability and reducing prototype rework.

Lamont-Doherty Earth Observatory — April 2021 – Aug 2021
Research Assistant — *Palisades, NY*

- Supported research on microplastic contamination in consumer detergents through sample preparation, filtration, **UV-Vis spectrophotometry**, and cleanroom laboratory testing.

Education

Northeastern University — Boston, MA
M.S. Mechanical Engineering, Concentration in Materials, Incoming Fall 2026 — *Sep 2026 – May 2028*

Rutgers University, School of Engineering — New Brunswick, NJ
B.S. Biomedical Engineering — *Sep 2021 – May 2025*

- Coursework in biomaterials, biomechanics, and biomedical devices focused on the mechanical, biological, and chemical behavior of engineered systems.
- Performed mechanical testing, failure analysis, and data analysis using **Instron** machines and optical microscopy.
- Conducted electronics lab coursework using **oscilloscopes** and **digital multimeters**, and ran FEA/simulation coursework in **SolidWorks**, **SolidWorks Simulation**, and **Abaqus**.

Projects

View full project portfolio: rkydin.github.io

UltraLight Sleeping Pad Pump — Nov 2025 – Present
Personal Project

- Designed and iterated a portable backpacking sleeping-pad pump through **three hardware revisions** in SolidWorks, engineering a sealed three-piece PETG housing with a centrifugal impeller to hit a **∼31 g** total weight under a sub-100 g ultralight requirement.
- Designed a custom **2-layer PCBA in KiCad** handling USB-C power input, overcurrent protection, motor kickback protection, and power filtering, replacing the off-the-shelf blower and trigger board used in earlier revisions.